Amazon Sales Data Analysis

1. Project Overview

This project analyzes Amazon Product Sales Analysis using Customer and Product data from 1465 Products and Reviews across various categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and average ratings to guide strategic business decisions.

2. Dataset Summary

- Rows: 1465 - Columns: 15 - Key Features:
- Customer demographics (User name, User ID)
- Product details (Product_name,Product_ID, Category, actual price, Product link ,Image link)
- Shopping behavior (Discount Applied, Rating, Review content, Review Rating, Rating count)
- Missing Data: values in Rating count column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.

- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_
count	1465	1465	1465	1465	1465	1465	1465	1463	
unique	1351	1337	211	550	449	92	28	1143	
top	B07JW9H4J1	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹999	50%	4.1	9,378	[CH FUN T cor
freq	3	5	233	53	120	56	244	9	

user_name	review_id	review_title	review_content	img_link	product_link
1465	1465	1465	1465	1465	1465
1194	1194	1194	1212	1412	1465
@\\TO|-|,Sethu madhav,Akash Thakur,Burger P...	R3F4T5TRYPTMIG,R3DQIEC603E7AY,R1O4Z15FD40PV5,R...	Worked on iPhone 7 and didn't work on XR,Good ...	I am not big on camera usage, personally. I wa...	https://m.media-amazon.com/images/I/413sCRKobN...	https://www.amazon.in/Wayona-Braided-WN3LG1-Sy...
10	10	10	8	3	1

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if Rating count has NULLs and dropped
- **Excel Integration:** Downloaded Cleaned Dataset from Python and loaded the cleaned DataFrame into Excel for Power bi Integration.

4. Data Analysis using Python (Business Questions)

We performed structured analysis in Python Pandas to answer key business questions:

```
[17]: print(df["product_name"].count())
```

```
1465
```

Total 1465 Products are there in the Amazon sale Data

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```
]: print(df["discount_percentage"].max())
```

```
94%
```

Highest Discount By a Product is 94%

```
9]: print(df["discount_percentage"].min())
```

```
0%
```

Lowest Discount By a Product is 0%

```
print(df["user_name"].value_counts())
```

```
user_name
$@|\|TO$|-|,Sethu madhav,Akash Thakur,Burger Planet,Justice ⚖️,indrajyoti d.,Aditya Kumar,E.C.GEORGE                    10
Manav,Adarsh gupta,Sundeep,S.Sayeed Ahmed,jaspreet singh,Khaja moin,Anand,S.ARUMUGAM                                   8
Satheesh Kadiam,Pritom Chakraborty,Vishwa,Simranpreet Singh,Saptarshi,Amazon Customer,D.RAGHUL,Dharmendra kumar         8
Omkar dhale,JD,HEMALATHA,Ajwadh a.,amar singh chouhan,Ravi Siddan,Himanshu Goel,Udaykumar                             7
ArdKn,Nirbhay kumar,Sagar Viswanathan,Asp,Placeholder,BharanI,sonia,Niam                                              7
                                                                                                                     ..
suraj,Bharanidhar,S.B.Vignesh,Srikanth GR,Hari,Shashi,Palanivel raja N,Pavan Singh                                     1
Bhuwaneshwar kathlane,Jainit mukeshbhai Chauhan,Mahesh prajapati,Sonu Kumar,narender,Rajkumar jain,Bani panda,Dhiraj Tiwari  1
lucky,SALMAN S,Rahul Singh,INSOLARE ENERGY PVT Ld,Sandeep Kumar Sahu,Amazon Customer,Sudhanshu,Jp                      1
Amazonnig,Anurag,p_v_,Anil Kumar,Ignatius Nayan D'Rozario,Amazon Customer,Maulik Aryan,basraj                          1
Rajib,Ajay B,Vikas Kahol,PARDEEP,Anindya Pramanik,Vikas Singh,Harshada Pimple,Saw a.                                   1
Name: count, Length: 1194, dtype: int64
```

There Are 1194 Customers Who Shoped Various Products From Amazon

1. **Top 5 Products by Rating** – Found products with the highest average review ratings.

```
]: df[['product_name', 'rating']].sort_values(by='rating', ascending=False).head(5)
```

]:

	product_name	rating
324	REDTECH USB-C to Lightning Cable 3.3FT, [Apple...	5.0
174	Syncwire LTG to USB Cable for Fast Charging Co...	5.0
775	Amazon Basics Wireless Mouse | 2.4 GHz Connect...	5.0
1145	Swiffer Instant Electric Water Heater Faucet T...	4.8
1201	Oratech Coffee Frother electric, milk frother ...	4.8

```
: df['rating'] = pd.to_numeric(df['rating'], errors='coerce')
```

```
: print(df['rating'].mean())
```

```
4.096584699453552
```

The Average Rating of All Customers is 4.096584699453552

	product_name	rating
324	REDTECH USB-C to Lightning Cable 3.3FT, [Apple...	5.0
174	Syncwire LTG to USB Cable for Fast Charging Co...	5.0
775	Amazon Basics Wireless Mouse | 2.4 GHz Connect...	5.0
1145	Swiffer Instant Electric Water Heater Faucet T...	4.8
1201	Oratech Coffee Frother electric, milk frother ...	4.8

Basic USBs have More Ratings

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2. **Top 3 Products per Category** – Listed the most purchased products within each category

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	about_product
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	Computers&Accessories|Accessories&Peripherals|...	₹399	₹1,099	64%	4.2	High Compatibility : Compatible With iPhone 12...
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹349	43%	4.0	Compatible with all Type C enabled devices, be...
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	Computers&Accessories|Accessories&Peripherals|...	₹199	₹1,899	90%	3.9	【Fast Charger& Data Sync】-With built-in safet...
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	Computers&Accessories|Accessories&Peripherals|...	₹329	₹699	53%	4.2	The boAt Deuce USB 300 2 in 1 cable is compati...

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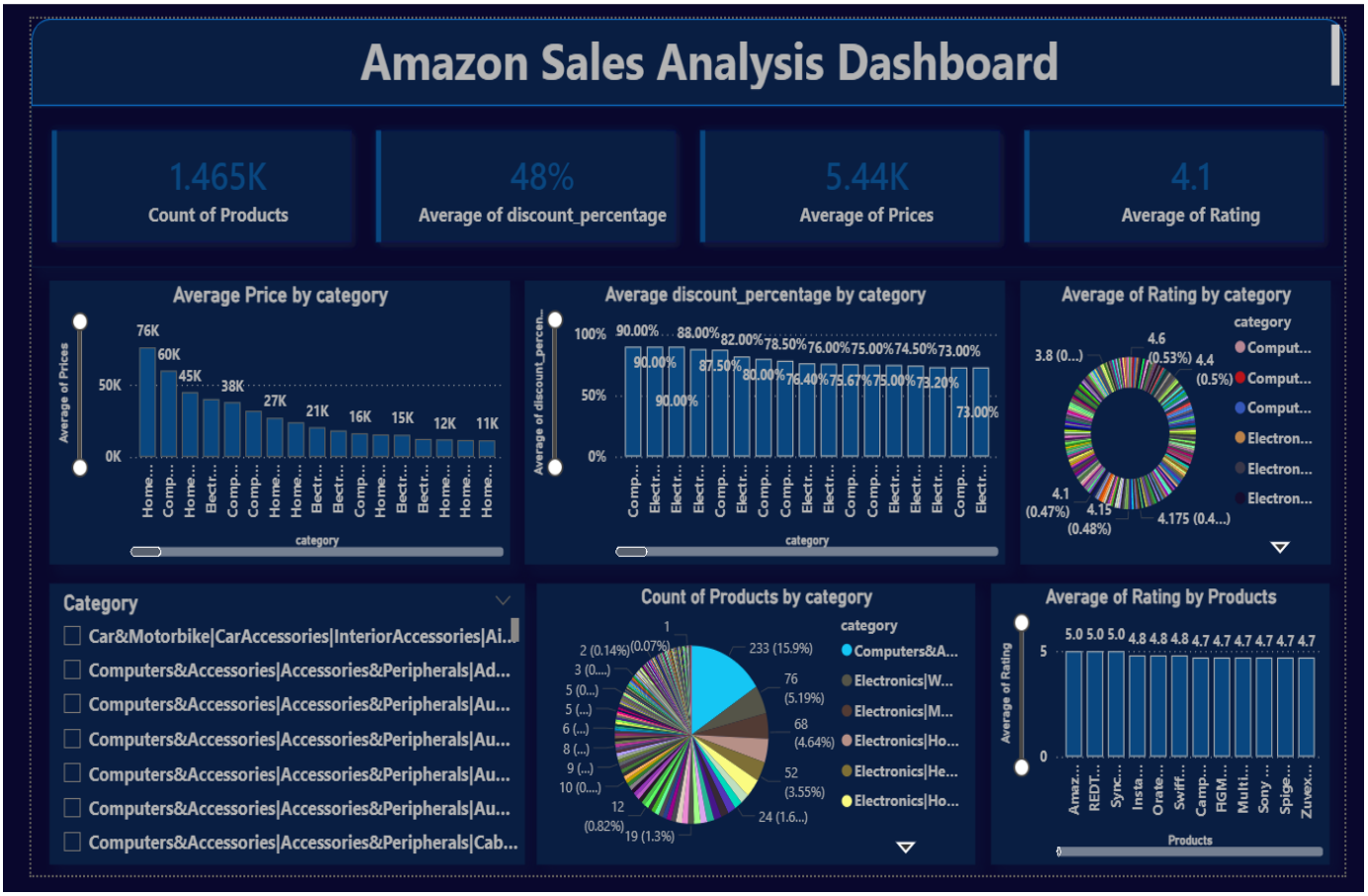
Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



Business Recommendations

- **Boost Discounts** – Promote exclusive benefits for Customers .
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the “Loyal” segment for Subscription Plans.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue Product Categories and express-shipping users.